

Folasewa Maryam Abdulsalam

Data Scientist

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EDUCATION

Msc. Neuroscience | GPA: 8/10

Sep 2025 - Present

Vrije Universiteit Amsterdam (VUA), Netherlands

Msc. Brain and Data Science | GPA: 94.44/100

Oct 2024 - Aug 2025

Bar-Ilan University (BIU), Israel

Bsc. Computer Science and Mathematics | GPA: 4.65/5.0 | Top 2/200

Apr 2017 - Aug 2023

Obafemi Awolowo University (OAU), Nigeria

TECHNICAL SKILLS

Languages & Frameworks	Python, MATLAB, TensorFlow, PyTorch
Neuro & Signal Tools	MNE-Python, EEGLAB, K-plan, Localite, COMSOL Multiphysics
Platforms	Visual Studio Code, Colab, FSL, Jupyter Notebook
Other	Git, LaTeX, ImageJ

EXPERIENCE

Graduate Researcher | EEG & Neurostimulation

Feb 2026 - Present

Neuroimaging Center (NIC), Mainz, Germany

- Designed and set up a concurrent TUS-EEG sleep experiment to study the role of thalamic sleep spindles in memory consolidation using transcranial ultrasonic stimulation (TUS).
- Simulated ultrasonic wave propagation through brain tissue in K-Plan to optimise targeting parameters for thalamic stimulation.
- Conducted a literature review on TUS parameter spaces and their neuromodulatory effects on thalamocortical circuits.

Graduate Researcher | Computational Neuromodulation

Dec 2024 - Jun 2025

Neuroengineering Lab, Bar-Ilan, Israel

- Built 2D axisymmetric finite-element models in COMSOL Multiphysics simulating microbubble-vessel interactions under focused ultrasound to identify acoustic parameters that maximize BBB permeability.
- Integrated fluid-structure interaction (FSI) and transient pressure acoustics to model vessel wall response across a parametric sweep of acoustic pressures and bubble radii.
- Abstract accepted for poster presentation at the Alzheimer's Association International Conference (AAIC), 2025.

Research & Innovation Intern | Machine Learning Research

Apr 2022 - Sep 2022

Data Science Nigeria, Lagos, Nigeria

- Performed systematic literature review on student engagement detection; delivered findings at IndabaX Nigeria.
- Analyzed call detail records for poverty mapping research.
- Delivered Python for Data Analysis training in an AI for Kids educational programme.

Junior Data Scientist | Neural Signal Analysis

May 2020 - Sep 2020

Korwave (Remote)

- Analyzed EEG recordings to build a seizure onset prediction pipeline for an epilepsy detection wearable targeting canine patients.
- Conducted literature review on state-of-the-art seizure detection algorithms; contributed to feature engineering and data labeling strategy for the ML model.

PROJECTS

EEG-Based Mental Workload Classification | [Project Link](#)

Mar 2026 - Apr 2026

Independent

- Implemented full EEG preprocessing and spectral feature extraction pipeline
- Implemented and compared NCA and Random Forest feature importance to identify spectral biomarkers
- Applied SHAP explainability to quantify which frequency bands and electrode locations drive classification performance.

Seizure Onset Prediction | [Project Link](#)

Mar 2025 - Apr 2025

Independent

- Replicated the Shoeb et al. (2010) spectral feature extraction pipeline using an 8-band filterbank (0.5–25 Hz) with 2-second sliding windows to capture preictal EEG dynamics.
- Trained and benchmarked logistic regression, random forest, and SVM classifiers to predict seizure onset.

EEG-based Alzheimer's Disease Classification | [Project Link](#)

Jan 2025 - Feb 2025

Independent

- Replicated multi-domain EEG feature extraction (spectral, complexity, connectivity) from Zheng et al. (2023) to capture Alzheimer's-related neurophysiological patterns.
- Trained and benchmarked SVM, Random Forest, and LightGBM classifiers to distinguish AD patients from healthy controls.

Student Engagement Detection System | [Project Link](#)

Jan 2023 - Jun 2023

Undergraduate Thesis

- Developed a YOLOv8-based engagement detection system trained on the DAiSEE dataset, providing student engagement feedback to lecturers.

EXTRACURRICULAR ACTIVITIES

Management, PulseRacing

Sept 2025 - Jan 2026

Demonstrator Lab, Vrije Universiteit Amsterdam

- Trained 2 individuals with spinal cord injury using Functional Electrical Stimulation (FES) Cycling;
- Scheduled interdepartmental meetings and monthly schedules.

Chairman, Final Year Senior Committee

Jan 2023 - Aug 2023

Department of Computer Science and Engineering, OAU

- Organized departmental final-year celebrations, overseeing a team of 7 specialized sub-committees covering budget, publicity, design, content, and logistics.

CONFERENCE PRESENTATIONS

Oral Presentation

Aug 2025

Deep Learning Indaba, Rwanda

AI-Augmented Simulation Framework for Optimizing Ultrasound-Mediated BBB Opening

Poster Presentation

Jul 2025

Alzheimer's Association International Conference (AAIC)

In Silico Modeling of Acoustic Microbubble Dynamics for Safe Drug Delivery

SELECTED AWARDS

- | | |
|---|-----------------|
| • Erasmus Mundus Joint Master's Scholarship (EU) 1 of 15 worldwide | Oct 2024 - 2026 |
| • First Runner-up, Code Cash Crop 3.0 Ag-Hackathon Farmer e-commerce platform | Aug 2022 |
| • Top 7 Finalist, AI4Energy Hackathon ML carbon footprint forecasting app | Sep 2021 |
| • MTN Foundation STEM Scholarship Top 500 STEM students nationally | Oct 2019 - 2023 |